Pair Trading Using Cointegration

Architecture & review brief: Nifty 50 classic pairs + ML (Phase 1) → Nifty 200 regime-aware relative-strength pairs (Phase 2) → Streamlit dashboard

Out of sample 2025-01 → 2026-10	Total return	Sharpe	Volatility	Max drawdown
Final strategy (point-in-time pairs)	**+6.8%**	**0.53**	**7.4%**	**−7.2%**
Nifty 200 TRI (1.5% dividend approx.)	−1.7%	0.00	14.2%	−14.6%
Goal met?	✔ higher	✔ higher	✔ lower	✔ smaller

Beta to the index 0.04. All returns net of NSE stock-futures costs + 5 bps slippage per side per leg.

1. The big picture

The repository contains two research phases and one dashboard that presents both. Phase 1 is the classic textbook approach and it failed out of sample; that failure motivated Phase 2, which is the main work and produced the final frozen strategy.

Phase 1 · notebooks/ + model_pipeline.py
22 Nifty 50 stocks in 4 sectors · OLS / Kalman hedge ratio β · spread z-score · Random Forest trade filter · realistic cost model · walk-forward · paper trading

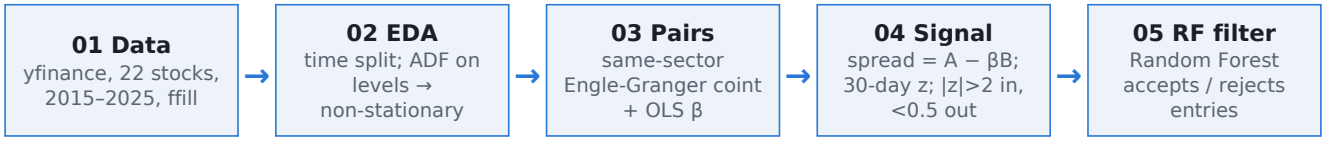
Phase 2 · root *.py scripts → research/
Full Nifty 200 with point-in-time membership · 1:1 log-ratio cointegration · relative-strength percentile signal · market-regime gate · fixed train/test protocol

Both feed **app/** (Streamlit multi-page dashboard), which reads the research outputs and a data bundle exported from the frozen final configuration.

Repository map

Area	Files	Role
Phase 1 notebooks	notebooks/01…05_*.ipynb	Data → EDA/ADF → pair selection → z-score backtest → RF filter
Phase 1 production	model_pipeline.py, costs.py, risk.py, simulate.py, walk_forward.py, paper_trading.py, dashboard.py	Feature engineering + per-pair RF models, cost card, sizing/stops, day-by-day engine, validation, live ledger
Phase 2 data	universe.py, pit_universe.py, tri.py, tools/parse_nse_*.py	Nifty 200 prices / index / VIX, point-in-time membership, TRI benchmark
Phase 2 regime	regime.py, kaufman_factors.py	Trend / breadth / volatility → 5 market states
Phase 2 engine	rs_pairs.py, perf.py	Pair formation, signals, portfolio backtester, costs, metrics
Phase 2 experiments	research.py, walk_forward_rs.py, tune_h15.py, kaufman_system.py, rs_lowvol_system.py	Successive designs (§3–§9 of REPORT.md)
Final strategy	pit_pair_system.py, pit_pair_robustness.py, final_oos.py	Frozen selection, OOS test, robustness
Outputs	research/ (CSV, JSON, figures, REPORT.md), data/, models/	All results and artefacts
Dashboard	app/streamlit_app.py, app/views/*, app/core/*, app/build_data.py	Presentation layer
Tests	tests/test_*.py (49 test functions)	No-lookahead, timing, cost, PIT and metric checks

2. Phase 1: classic cointegration pairs + ML filter



- **Split:** train 2015–2021, validation 2022–2023, test 2024–2025 (time-ordered, no shuffling).
- **Selected pairs:** HEROMOTOCO/EICHERMOT (Auto), HDFCBANK/KOTAKBANK (Banking), INFY/HCLTECH (IT); all cointegrated at 5% (p ≈ 0.0006–0.0075).
- **RF features:** z, z_chg_5, z_chg_20, corr_30, spread_vol_20, spread_vol_60 (300 trees, depth 6, balanced class weights). model_pipeline.py adds a Kalman / recursive-least-squares β, half-life, Hurst exponent and β deviation, and trains one model per pair (models/pair_models/*.joblib).

Production layer

Module	What it does
costs.py	Indian cash-equity rate card: STT 0.1% both sides, exchange 0.00297% + 18% GST, SEBI fee, stamp duty 0.015% (buy), DP charge, SLB borrow 8%/yr on the short leg, volatility-scaled slippage, integer share rounding.
risk.py	Volatility-target sizing (1% daily risk), z-stop at 4, 40-day maximum hold, portfolio limits.
simulate.py	Pure state-in / state-out step(): open, hold or close one pair for one day using real share counts.
walk_forward.py	Expanding-window retraining (4-year start, 6-month steps), chained out-of-sample equity curve.
paper_trading.py	Run once per trading day; persists positions, trades and equity to CSV; idempotent re-runs.
dashboard.py	Original Streamlit app (pair explorer, baseline vs ML, risk/costs, live signals).

Phase 1 results

Strategy	CAGR	Sharpe	Max DD
Baseline z-score	0.7%	0.14	−42.9%
RF-filtered	−14.5%	−1.03	−78.2%
Walk-forward portfolio (3 pairs)	−2.1%	−1.00	−12.0%

Takeaway for the review: the ML filter did **not** help; it made performance worse. Present Phase 1 as the baseline experiment whose failure (few pairs, unstable regression hedge, cash-market costs, no regime awareness) motivated the Phase 2 redesign.

3. Phase 2: system architecture



3.1 Data layer

- **universe.py:** adjusted daily closes for the Nifty 200, the ^CNX200 index and India VIX from 2012 (Yahoo Finance). NSE's Industry column (19 industries) is the sector grouping.
- **pit_universe.py:** reconstructs **point-in-time membership**: official month-end constituent PDFs (Dec 2013–Mar 2022) plus NSE "Replacements in indices" press releases (2022–2026). 420 stocks were members at some point; a rename map handles symbol changes (ZOMATO→ETERNAL, LTI/LTIM→LTM, ...). Removes survivorship bias.
- **tri.py / final_oos.nifty200_tri:** benchmark is the Nifty 200 Total Return Index. NSE's endpoint is bot-protected, so it is approximated as price index + 1.5%/yr dividends reinvested daily (at or above the real yield, so harder to beat). An official file in data/nifty200_tri.csv is used automatically if present.

3.2 Regime engine

Five states, first match wins: **HIGH_VOL → BULLISH → BEARISH → LOW_VOL → SIDEWAYS**. Everything is trailing, so the label for day t is known at t's close and only gates trades from t+1.

	regime.py (v1)	kaufman_factors.py (v2, used in final model)
Trend	Distance from 21-day EMA	Kaufman Efficiency Ratio (signed, 20d), ROC 63d, EMA distances
Breadth	% above 200 EMA, A/D ratio 10d	Net new 52-week highs, A/D ratio, % above 50/200 EMA
Volatility	VIX + realised vol percentiles	VIX, realised vol, ATR proxy, cross-sectional dispersion
Combination	+1/−1 votes; ≥ +2 Bullish, ≤ −2 Bearish	Trailing percentile ranks → greedy selection by rank-IC with $\|\rho\| \le 0.70$ → mean direction score and vol score

Final regime factors: direction = A/D ratio, Efficiency Ratio, net new highs, ROC63; volatility = VIX, ATR proxy, dispersion. 756-day percentile window, bull/bear thresholds 0.55/0.45, high-vol ≥ 0.90.

3.3 Core engine: rs_pairs.py

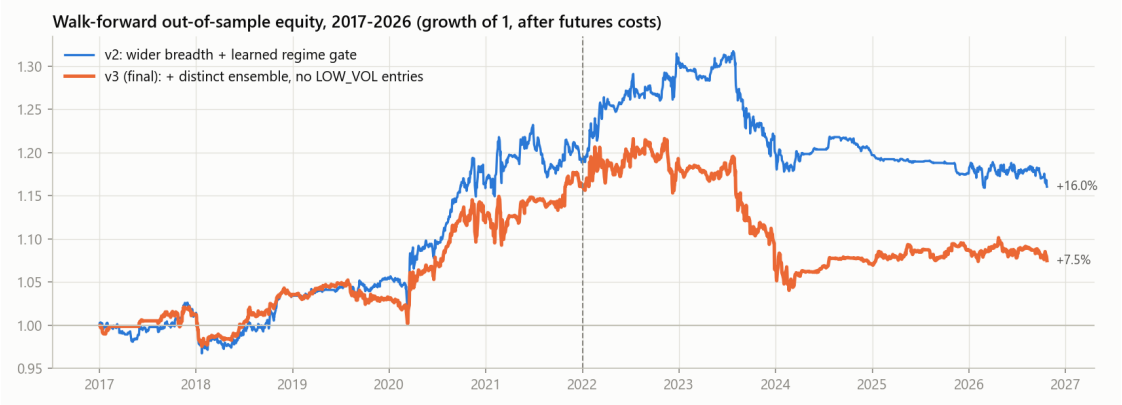
Stage	Logic
Pair formation	Every January and July, test all same-industry pairs on the trailing window only. ADF on ln(A/B) = cointegration with a **fixed 1:1 hedge** (no regression β). Filters: ADF p ≤ 0.05, return correlation ≥ 0.5, no daily move > 40% (bad-data guard). Pairs trade only in the next 6 months, so selection is out-of-sample by construction.
Signal	Relative strength $RS_L = \ln(A/B)_t - \ln(A/B)_{t-L}$. Variants: rs_vol (RS / ratio vol), **rs_pct** (percentile of RS in a trailing window, used in final), stress (Kaufman stress indicator).
Daily loop	1) mark to market → 2) exits (time, revert, stop, missing price, end of test) → 3) fill yesterday's entry decisions at today's close (1-day lag) → 4) rank new signals by |score| and fill free slots.
Portfolio	Short the outperformer, buy the laggard, equal rupee legs (equity / max_pairs). Slots per regime (0 = blocked), max_per_stock cap.
Costs	futures: STT 0.02% sell side, exchange + GST, stamp, SEBI, monthly roll, no borrow. cash: costs.py card + 8%/yr borrow. Plus 5 bps slippage per side.

4. Research journey: what was tried and why it was replaced

REPORT §	Script	Design	Outcome
§3–4.1	research.py	Static grid tuned on 2014–21, holdout 2022+	In-sample Sharpe 1.11 → holdout −0.56. Overfit to the 2020–21 COVID dispersion.
§4.2	walk_forward_rs.py	Yearly walk-forward ensemble of top-5 configs + regime gate	OOS 2017–2026 Sharpe 0.19, beta 0.02. Positive but t ≈ 0.6, not significant.
§7	tune_h15.py	15-day hold, every lookback tuned walk-forward	No positive edge; 3-year cointegration window most robust; re-tuning regime windows chases noise.
§8	kaufman_system.py	Regime-timed index sleeve + pair sleeve, train 2015–24 / test 2025+	Train Sharpe 1.00 → test −0.60. Index timing whipsawed.
§9	rs_lowvol_system.py	RS + low-vol stock selection + hedge + pairs	Test failed on all four criteria vs both benchmarks.
§10	**pit_pair_system.py**	**Point-in-time universe, pairs only, fixed protocol**	**All four criteria met vs TRI out of sample.**

Key empirical findings carried forward

- Mean reversion is the right direction: the momentum variant had Sharpe −1.31 on development data.
- Cointegration adds value over plain correlation (correlated non-cointegrated control: 0.49 vs 0.72).
- Hold of 1–2 days loses to costs everywhere; the good zone is 10–20 days.
- Pair reversion works in **Sideways** and **High Vol** markets and fails in **Low Vol**, consistent with reversal profits being a payment for liquidity provision (Nagel 2012).
- Survivorship mattered: the §8 pair sleeve made +0.9% on today's list but −2.8% on actual point-in-time members.



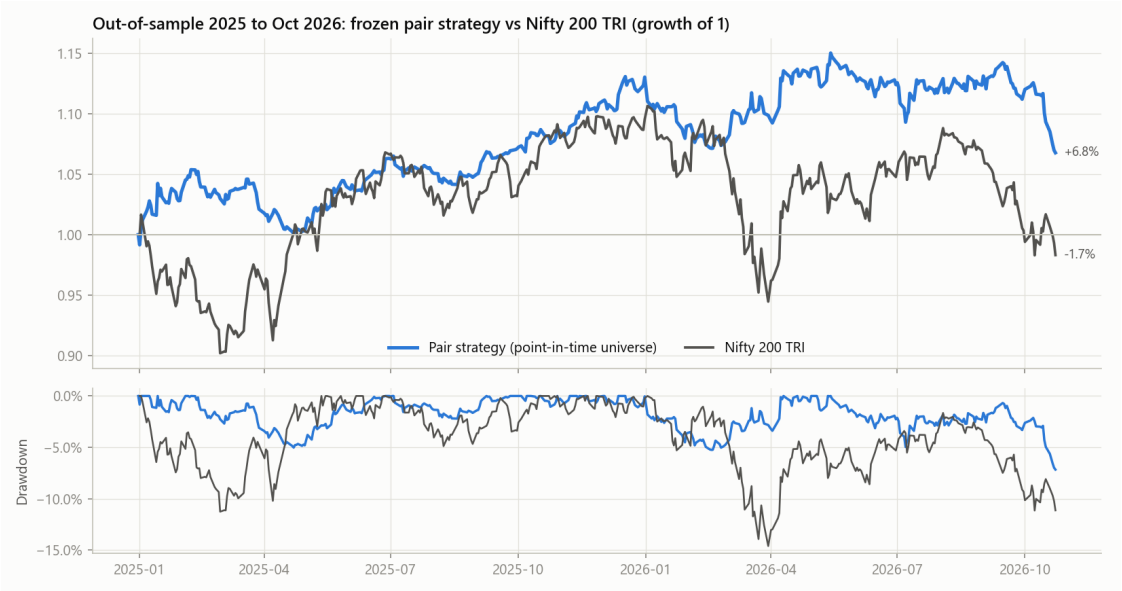
Walk-forward (§4.2) out-of-sample equity, 2017–2026.

5. The final frozen model (§10)

Train
2015–2024 only
→
Stage 1: regime
1,080 candidates, scored + plateau-smoothed
→
Stage 2: pairs
72 signal settings × 3 regime modes, ≥ 100 trades
→
Freeze
research/pit_pairs/final_config.json
→
Test once
2025-01 → 2026-10 vs TRI

Scoring (Kaufman-style robustness, no z-scores): percentile-rank composite of min-half Sharpe (2015–19 vs 2020–24), min-half MAR and full Sharpe; then **plateau smoothing**, i.e. each candidate's score is averaged with its grid neighbours so a broad stable region beats a lucky isolated spike.

Parameter	Selected value
Signal	Percentile rank of 20-day relative strength within a 126-day window (rs_pct, L = 20)
Entry / exit	Enter at the 90th / 10th percentile; exit when RS crosses its median, max 15 calendar days
Pair formation	3-year (756-day) cointegration window, same industry, point-in-time members only
Regime gate	Trade in every regime except Low Vol
Book	10 pair slots, a stock in at most 2 pairs, equal rupee legs
Costs	NSE stock-futures charges + 5 bps slippage per side per leg



Out-of-sample equity: point-in-time pair strategy vs Nifty 200 TRI.

Results and robustness

Period / check	Return	Sharpe	Vol	Max DD	Criteria
OOS 2025 → 2026-10	**+6.8%**	**0.53**	**7.4%**	**−7.2%**	**4/4**
Train 2015–2024	+13.8%	0.21	8.0%	−20.6%	2/4
OOS, slippage 10 bps	+0.4%	0.07	–	–	4/4 (narrowly)
OOS, slippage 20 bps	−11.3%	−0.86	–	–	1/4
OOS, cash + 8% borrow	−13.3%	−1.04	–	–	1/4
OOS, TRI at 2.5% dividends	+6.8%	0.53	–	–	4/4

By year: 2025 +13.0% vs TRI +10.0%; 2026 YTD −5.5% vs −10.2%. Of the top-20 training settings, 9 meet all four criteria out of sample, exactly those using the 3-year cointegration window.

6. Dashboard and tests

Streamlit dashboard (app/)

- **streamlit_app.py:** multi-page navigation: Overview, How the model works, Data & PIT universe, Regime engine, Final results, All pairs & trades, Pair explorer, Live signal monitor, Phase 1, Phase 2, Limitations.
- **core/data.py** cached loaders · **core/charts.py** Plotly figures · **core/style.py** CSS theme.
- **build_data.py** re-runs the frozen configuration, **asserts it reproduces research/pit_pairs/full_equity.csv exactly**, and exports parquet/CSV to app/data/. The deployed app then needs only streamlit, pandas, numpy, plotly and pyarrow (app/requirements.txt).
- Deployable on Streamlit Community Cloud (main file app/streamlit_app.py); .devcontainer for Codespaces.

Test suite (tests/)

File	Guarantees checked
test_rs_pairs.py	No lookahead in scores or regimes; next-close fills and exact holding period; 15-calendar-day exit over weekends/holidays; formation uses only pre-period data; net = gross − costs; equity = sum of trade P&L; slot and per-stock caps; regime blocking; vol filter; stop and revert exits; metrics against known values.
test_kaufman.py	Efficiency Ratio bounds; percentile normalisation without lookahead; stress indicator; whipsaw confirmation; state priority; correlated / wrong-sign factor removal; index sleeve 2-day lag.
test_pit.py	200 members at each review; reviews applied on effective date; renames; member mask; TRI approximation and scorecard.
test_rs_lowvol.py	No future data in targets; selection logic; turnover buffer; hedge lag and cost; criteria.

7. How to reproduce

```
python universe.py → python regime.py → python pit_pair_system.py → python pit_pair_robustness.py → python report_figures.py → python -m pytest tests -q → python app/build_data.py → streamlit run app/streamlit_app.py
```

8. Likely review questions

Q. Why a fixed 1:1 hedge instead of a regression β?

A fixed hedge cannot be overfit. A stationary ln(A/B) means the two rupee values are cointegrated. Rolling OLS / Kalman β (Phase 1) adds estimation noise and was part of why Phase 1 was unstable.

Q. How is lookahead bias avoided?

Every indicator is trailing; decisions at t's close fill at t+1's close; pairs are formed only on data before each period; percentile windows include only past days. Unit tests enforce each of these.

Q. What about survivorship bias?

Handled with the point-in-time universe. It mattered: +0.9% on today's list vs −2.8% on actual members for the earlier sleeve. Residual: 46 delisted / merged names have no price data.

Q. Isn't this overfit?

Selection used only 2015–2024, with half-period consistency, plateau smoothing and a single test run. Honest caveat: the 2025–26 window had already been seen by the §8–§9 designs, so it is not a perfectly clean holdout.

Q. Is the result statistically significant?

No. A 22-month Sharpe of 0.53 has a standard error of about 0.75. Training Sharpe was only 0.21 and the strategy lagged the TRI on return in 9 of 10 training years; its edge is mainly lower risk (vol, drawdown, beta ≈ 0).

Q. Why must it be traded through futures?

Cash-delivery STT on both sides plus stock-borrow cost removes the edge (−13.3% OOS). The backtest assumes all names are F&O-tradable, which could not be verified.

Q. Why skip Low Vol?

Low Vol was negative in both development and holdout studies; reversal profits pay for liquidity, which earns least in calm markets (Nagel 2012).

Q. Why percentiles instead of z-scores?

Percentile ranks are bounded, robust to fat tails and adapt to changing levels; the brief for Phase 2 was to avoid z-scores entirely.

Q. What is the next step?

Forward paper trading of the frozen configuration, the official TRI and F&O list, and a longer multi-year test.

9. Inconsistencies a reviewer may spot

- README describes the Phase 1 RF filter as reducing drawdown, but data/ml_vs_baseline_stats.csv shows it worsened results (Sharpe −1.03, max DD −78%).
- Test counts differ: README says 58, REPORT §6 says 24; the code has 49 test functions (one parametrised).
- pit_pair_system.py docstring says 47 members lack price data; REPORT says 46.
- The README's mermaid diagram shows only the Phase 1 pipeline, not the final Phase 2 system.